

Legendre Polynomials and Functions

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- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Introduction

Legendre's differential equation arises from separation of variables in spherical coordinates. For example, the azimuthally symmetric angular equation is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0.$$

Set $x = \cos \theta$ and write $\Theta(\theta) = y(x)$. The chain rule gives every step in the change of variable:

$$\begin{aligned} \frac{d\Theta}{d\theta} &= \frac{dy}{dx} \frac{dx}{d\theta} = -\sin \theta y', \\ \sin \theta \frac{d\Theta}{d\theta} &= -(1 - x^2)y', \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) &= -\frac{d}{dx} [-(1 - x^2)y'] \\ &= (1 - x^2)y'' - 2xy'. \end{aligned}$$

Introduction (cont.)

Requiring a nonzero solution that remains bounded at both poles $x = \pm 1$ turns this singular Sturm–Liouville problem into an eigenvalue problem. Its eigenvalues are $\lambda = n(n+1)$, $n = 0, 1, 2, \dots$, and hence

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0. \quad (\text{C.1.1})$$

On an open interval not containing the singular endpoints, two linearly independent solutions are the Legendre functions of the first and second kinds, denoted by $P_n(x)$ and $Q_n(x)$. For nonnegative integer n , P_n is the degree- n solution that is finite at both endpoints, whereas Q_n is singular there.

The polynomial $P_n(x)$ is defined for every real or complex x ; the interval $[-1, 1]$ is its angular and orthogonality domain, not the domain of the polynomial itself. These ordinary Legendre polynomials express the polar-angle dependence of azimuthally symmetric separated solutions of Laplace's and Helmholtz equations; nonzero azimuthal order uses the associated functions introduced below. Exact spherical symmetry would retain only $n = 0$; general spherical-coordinate problems need the higher angular orders.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Legendre Polynomials

The first few Legendre polynomials are

$$P_0(x) = 1,$$

$$P_1(x) = x,$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1),$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x),$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3),$$

$$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x). \quad (\text{C.2.1})$$

They follow from Rodrigues' formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n, \quad n = 0, 1, 2, \dots \quad (\text{C.2.2})$$

Parity, endpoint values, and recurrence

The generating function is

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad (C.2.3)$$
$$|t| < 1, \quad -1 \leq x \leq 1.$$

It makes the parity and endpoint values transparent:

$$\begin{aligned} g(-x, t) &= g(x, -t) \\ &= \sum_{n=0}^{\infty} (-1)^n P_n(x) t^n, \end{aligned} \quad P_n(-x) = (-1)^n P_n(x),$$

$$g(1, t) = (1 - t)^{-1} = \sum_{n=0}^{\infty} t^n, \quad P_n(1) = 1,$$

$$\begin{aligned} g(-1, t) &= (1 + t)^{-1} \\ &= \sum_{n=0}^{\infty} (-1)^n t^n, \end{aligned} \quad P_n(-1) = (-1)^n.$$

Parity, endpoint values, and recurrence (cont.)

In particular, $P_n(0) = 0$ for odd n . Differentiating (C.2.3) with respect to t gives

$$(1 - 2xt + t^2) \frac{\partial g}{\partial t} = (x - t)g.$$

Substitution of $g = \sum_{n \geq 0} P_n t^n$ and comparison of the coefficient of t^n yields

$$(n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1} = xP_n - P_{n-1},$$

and therefore

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \quad n \geq 1. \quad (\text{C.2.4})$$

Orthogonality and normalization

Equation (C.1.1) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) = 0. \quad (\text{C.2.5})$$

Write this equation once for P_n , once for P_m , multiply by P_m and P_n , respectively, and subtract. The product rule then gives

$$\begin{aligned} \frac{d}{dx} \{ (1-x^2) [P_m P'_n - P_n P'_m] \} \\ + [n(n+1) - m(m+1)] P_m P_n = 0. \end{aligned}$$

After integration from -1 to 1 , the boundary term vanishes because P_n, P_m and their derivatives are finite while $1-x^2=0$ at both endpoints. For $m \neq n$, the eigenvalues are distinct, so

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0.$$

Orthogonality and normalization (cont.)

For $m = n$, Rodrigues' formula supplies the norm. Integrating by parts n times (all boundary terms vanish because $(x^2 - 1)^n$ has zeros of order n at $x = \pm 1$) gives

$$\begin{aligned}\int_{-1}^1 P_n^2 dx &= \frac{(-1)^n}{2^n n!} \int_{-1}^1 P_n^{(n)}(x)(x^2 - 1)^n dx \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \int_{-1}^1 (1 - x^2)^n dx.\end{aligned}$$

Here Rodrigues' formula gives $P_n^{(n)} = (2n)!/(2^n n!)$. The beta integral evaluates to

$$\int_{-1}^1 (1 - x^2)^n dx = \frac{2^{2n+1}(n!)^2}{(2n+1)!},$$

so the complete orthogonality and normalization relation, with δ_{mn} the Kronecker delta, is

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}, \quad m, n = 0, 1, 2, \dots \quad (\text{C.2.6})$$

Orthogonality and normalization (cont.)

Consequently, if $f \in L^2[-1, 1]$, its Legendre-series coefficients are obtained by multiplying the expansion by P_m and using (C.2.6):

$$f(x) \sim \sum_{n=0}^{\infty} a_n P_n(x), \quad a_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx. \quad (\text{C.2.7})$$

Pointwise convergence requires the usual additional regularity conditions; square-integrability guarantees convergence in the L^2 sense.

Orthogonality and normalization

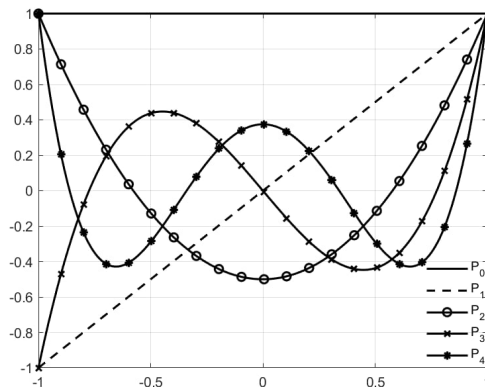


Figure C.1: Legendre polynomials for $n = 0, 1, 2, 3, 4$.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions**
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Associated Legendre Functions

For $-1 \leq x \leq 1$, this book uses the phase-free engineering convention

$$P_n^m(x) = (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad 0 \leq m \leq n. \quad (\text{C.3.1})$$

Many mathematical-physics references include the Condon–Shortley factor $(-1)^m$. Thus the convention here and that convention are related by

$$P_{n,\text{here}}^m(x) = (-1)^m P_{n,\text{CS}}^m(x).$$

No differential equation or squared normalization changes under this constant phase. In particular, the convention used here gives the identity employed in Chapter 4,

$$P_n^1(\cos \theta) = \sin \theta P_n'(\cos \theta) = -\frac{d}{d\theta} P_n(\cos \theta).$$

To derive the associated equation without skipping the differentiation step, let $u = d^m P_n / dx^m$. Differentiating (C.1.1) m times and applying the product rule gives

$$(1 - x^2)u'' - 2(m + 1)xu' + [n(n + 1) - m(m + 1)]u = 0. \quad (\text{C.3.2})$$

Associated Legendre Functions (cont.)

Now put $s = 1 - x^2$ and $v = s^{m/2}u = P_n^m$, so that

$$\begin{aligned}u &= s^{-m/2}v, \\u' &= s^{-m/2} \left(v' + \frac{mx}{s}v \right), \\u'' &= s^{-m/2} \left[v'' + \frac{2mx}{s}v' + \frac{m(1 + (m+1)x^2)}{s^2}v \right].\end{aligned}$$

Substitution into (C.3.2) and collection of the v' and v terms gives

$$(1 - x^2)v'' - 2xv' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] v = 0. \quad (\text{C.3.3})$$

With $x = \cos \theta$, the regular polar factor is $P_n^m(\cos \theta)$. For $m > 0$, multiplication by either real factor $\cos(m\phi)$ or $\sin(m\phi)$, or equivalently use of the complex pair $e^{\pm im\phi}$, supplies the azimuthal dependence. For $m = 0$, the sole nonzero azimuthal factor is the constant 1; $\sin(0\phi)$ is identically zero and is not a second solution.

Associated Legendre Functions (cont.)

For fixed m , equation (C.3.3) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)v_n'] + \left[n(n+1) - \frac{m^2}{1-x^2} \right] v_n = 0, \quad v_n = P_n^m.$$

Write this equation for v_n and v_l , multiply by v_l and v_n , and subtract:

$$\frac{d}{dx} \{ (1-x^2) [v_l v_n' - v_n v_l'] \} + [n(n+1) - l(l+1)] v_l v_n = 0.$$

After integration on $[-1, 1]$, the boundary term vanishes. For $m = 0$ this is the endpoint argument already used for P_n ; for $m \geq 1$, $v_n = (1-x^2)^{m/2}$ times a polynomial, so each boundary product is $O((1-x^2)^m)$. Thus distinct degrees are orthogonal for fixed m . The norm follows by integrating by parts m times:

$$\begin{aligned} \int_{-1}^1 [P_n^m(x)]^2 dx &= \int_{-1}^1 (1-x^2)^m [P_n^{(m)}(x)]^2 dx \\ &= (-1)^m \int_{-1}^1 P_n(x) \end{aligned}$$

Associated Legendre Functions (cont.)

$$\times \frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] dx.$$

Differentiating Rodrigues' formula gives

$$\frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] = (-1)^m \frac{(n+m)!}{(n-m)!} P_n(x).$$

Using (C.2.6), one obtains

$$\int_{-1}^1 P_l^m(x) P_n^m(x) dx = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} \delta_{ln}, \quad (\text{C.3.4})$$

for $l, n \geq m \geq 0$. Equivalently, $x = \cos \theta$ and $dx = -\sin \theta d\theta$ give

$$\int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}. \quad (\text{C.3.5})$$

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind**
- 5 Summary of Important Relations

Legendre Functions of the Second Kind

On $-1 < x < 1$, the real second solution (more precisely, the Ferrers function of the second kind) is

$$\begin{aligned} Q_n(x) = & \frac{1}{2} P_n(x) \ln\left(\frac{1+x}{1-x}\right) \\ & - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \\ & n = 0, 1, 2, \dots \end{aligned} \tag{C.4.1}$$

where the sum is empty for $n = 0$. For example,

$$\begin{aligned} Q_0(x) &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \\ Q_1(x) &= \frac{x}{2} \ln\left(\frac{1+x}{1-x}\right) - 1, \\ Q_2(x) &= \frac{3x^2-1}{4} \ln\left(\frac{1+x}{1-x}\right) - \frac{3x}{2}. \end{aligned}$$

Legendre Functions of the Second Kind (cont.)

Direct substitution verifies (C.1.1); alternatively, these functions follow from Q_0, Q_1 using the same three-term recurrence (C.2.4). To retain the Wronskian sign, divide the differential equation by $1 - x^2$ and set $W = P_n Q'_n - P'_n Q_n$. Abel's identity gives

$$W' = \frac{2x}{1-x^2} W \implies W = \frac{C}{1-x^2}.$$

As $x \rightarrow 1^-$, $P_n \rightarrow 1$ and the leading logarithmic term in Q_n gives $(1-x^2)Q'_n \rightarrow 1$, whereas $(1-x^2)P'_n Q_n \rightarrow 0$; hence $C = 1$. Therefore

$$P_n Q'_n - Q_n P'_n = \frac{1}{1-x^2},$$

so P_n and Q_n are linearly independent on $(-1, 1)$. The logarithm makes Q_n singular at $x = \pm 1$.

The real Legendre function on either exterior interval $|x| > 1$ uses a different real logarithmic expression,

$$\begin{aligned} Q_n(x) = & \frac{1}{2} P_n(x) \ln\left(\frac{x+1}{x-1}\right) \\ & - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \quad |x| > 1. \end{aligned} \tag{C.4.2}$$

Legendre Functions of the Second Kind (cont.)

for which the logarithm's ratio is positive. Complex analytic continuation requires a specified branch and must not be obtained by naively inserting $|x| > 1$ into the interior principal-log formula. The exterior $Q_n(\xi)$, with ξ the spheroidal radial coordinate, occurs, for example, as a radial factor in prolate-spheroidal separations. For a spherical exterior problem, by contrast, the radial Laplace factors are $r^{-(n+1)}$, not $Q_n(\cos \theta)$.

Table of contents

- 1 Introduction
- 2 Legendre Polynomials
- 3 Associated Legendre Functions
- 4 Legendre Functions of the Second Kind
- 5 Summary of Important Relations

Summary of Important Relations

For $n = 0, 1, 2, \dots$, and $0 \leq m \leq n$, the principal relations are

$$\begin{aligned}\frac{1}{\sqrt{1-2xt+t^2}} &= \sum_{n=0}^{\infty} P_n(x)t^n, \\ |t| < 1, \quad -1 \leq x \leq 1, \\ \int_{-1}^1 P_m(x)P_n(x) dx &= \frac{2}{2n+1} \delta_{mn}, \\ \frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) &= 0, \\ \int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta \\ &= \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}.\end{aligned}$$